Governing Datacenter Growth: Regulatory Challenges, Technical Constraints, and Policy Responses Across States

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Abstract

The rapid expansion of hyperscale data centers is creating a new class of public-policy problems at the intersection of electric utility regulation, environmental governance, and computing-infrastructure design. This paper asks how states should govern data center integration into electricity systems and local resource environments given the facilities' distinctive technical characteristics: concentrated and rapidly increasing electric demand, high reliability requirements, backup-power coordination needs, and cooling-system tradeoffs. The paper compares emerging state approaches in Texas, Minnesota, South Dakota, New York, and Virginia. The analysis finds that the most technically grounded policies do not treat data centers as passive industrial or commercial facilities. They create formal utility–operator interfaces around interconnection readiness, load commitments, flexibility options, stranded-cost protection, clean-energy alignment, water availability, cooling design, and environmental performance. Texas provides the strongest enacted model for interconnection discipline and emergency grid operations; Minnesota links large-customer rates, clean-energy procurement, water permitting, and sustainable design; South Dakota emphasizes direct cost responsibility, water reporting, and local authority; New York has adopted a precautionary environmental-review model while permanent rules are developed; and Virginia combines voluntary flexibility, electricity-use taxation, water-efficient cooling requirements, water reporting, and backup-generator emissions controls. The paper concludes that durable data center governance requires explicit rules for cost sharing, transparency, and environmental performance, not ad hoc negotiation alone.

1 Introduction and Background

Data centers have become a central form of industrial infrastructure for cloud computing, artificial intelligence, content delivery, enterprise information technology, and large-scale analytics. They are not merely passive warehouses for stored data. A modern facility combines dense server fleets, high-bandwidth networking, power-conversion equipment, backup generation, cooling systems, and security infrastructure into a continuously operating compute plant. For public-policy purposes, this means that data centers are best understood as large, persistent electric loads with associated land, water, air-quality, and local-infrastructure effects.

The recent policy urgency comes from the scale and speed of projected load growth. Large data centers are commonly measured in megawatts (MW) because electricity demand is the operational feature that most directly affects utility planning, interconnection queues, transmission investment, and emergency grid operations. The 2024 United States Data Center Energy Usage Report prepared for the U.S. Department of Energy estimated that data centers accounted for roughly 3 percent of U.S. electricity use in 2024 and could account for as much as 12 percent by 2028 under high-growth scenarios [16]. That growth is driven not only by traditional cloud services but also by artificial-intelligence workloads, which can require dense specialized computing hardware and can create sharper changes in demand than many conventional industrial processes.

Those characteristics create a difficult regulatory problem. Data centers can bring construction activity, tax revenue, improved telecommunications infrastructure, and economic-development opportunities. Federal and national-security discussions increasingly frame data centers and cloud infrastructure as strategic assets for artificial intelligence, scientific leadership, and economic competitiveness [28]. Yet the speed of data center growth, and the sector's large footprint on electricity systems and water resources, are creating substantial concerns about its economic and environmental impacts. New substations and transmission lines, higher peak demand, exposure of ordinary ratepayers to stranded infrastructure costs, water withdrawals or consumptive cooling, diesel or gas backup-generator emissions, land-use conflict, noise, and cumulative effects when multiple facilities cluster in the same community all show why public regulation can help mitigate the downsides of data center development.

Data center regulation, however, sits at the intersection of federal, state, and municipal authority. Federal permitting, environmental law, energy policy, and national-security priorities shape the outer context; state utility, water, environmental, and tax regimes determine many of the binding conditions for service and development; and municipal land-use, zoning, noise, and infrastructure decisions often decide how a project is experienced on the ground. Together, these rules determine how and where data centers can be built, making it difficult for any single public body to address all economic and environmental concerns on its own. The structure of electricity regulation makes the state level especially important. The Federal Energy Regulatory Commission (FERC) regulates interstate electric transmission and wholesale electricity sales, but retail electricity sales, local distribution, many facility-siting decisions, and customer-facing utility rates are principally state or local responsibilities [6, 10]. As a result, the most pressing data center questions about large-load service terms, ratepayer protection, utility cost recovery, state water permits, environmental conditions, and tax treatment often run through state institutions: utility commissions, water-permitting agencies, environmental regulators, state tax authorities, and state legislatures.

The policy challenge is therefore not concerned with defining the value of data centers to the broader economy or individual consumers. The challenge is how states can allow useful digital infrastructure to develop while making sure that the facilities internalize the system costs they create, disclose enough information for regulators and communities to evaluate impacts, and operate flexibly where doing so can protect the electric grid as well as ratepayers. This is why the most consequential state policies tend to converge around a common set of tools: large-load interconnection rules, project-readiness requirements, special tariffs or customer classes, demand-response and emergency-curtailment mechanisms, clean-energy or efficiency obligations, water-use review and reporting, air-emissions standards for backup generators, and siting or environmental-review processes.

This paper uses that policy context to compare five leading state approaches. The comparison is not intended to rank states by how welcoming they are to data center development, nor by the number of facilities they host. It instead asks which policy designs are technically coherent: which ones recognize that data centers have distinctive load shapes, interconnection behavior, backup-power capabilities, cooling architectures, and reliability expectations, and which ones translate those characteristics into enforceable public obligations.

2 Technical Governance Problem

The central governance mismatch is that modern data centers do not fit comfortably within rules designed for ordinary commercial or industrial facilities. Their unusually large and rapidly developing electric loads affect grid planning and costs; their backup-generation and energy-sourcing choices affect emissions; and their cooling systems can impose significant electricity and water demands. These effects are technically interconnected: measures that reduce water consumption may increase electricity use, while backup resources that improve reliability may increase local air emissions. Effective governance must therefore jointly address these concerns.

Cooling technology illustrates these cross-resource tradeoffs. Computing equipment converts much of the electricity it consumes into heat, which cooling systems must remove using additional electricity to operate chillers, pumps, fans, and other heat-rejection equipment. The cooling architecture therefore affects both the facility's total electric demand and its water demand: water-intensive evaporative systems can reduce the electricity needed for heat rejection under some conditions, while dry or other water-conserving systems may require more electricity to provide equivalent cooling. Federal energy-management guidance describes water-usage effectiveness as a site-based metric tied to IT energy use and identifies operational and technology options for reducing cooling-water demand [22]. Congressional Research Service analysis similarly emphasizes that centralized cooling systems can rely on chilled air or chilled-water loops and that water-consuming heat rejection can materially affect local water demand [5]. For policy purposes, cooling choices therefore determine whether the burdens of data center growth fall on local water supplies, electric systems, or both.

Data center load is also flexible rather than a single undifferentiated block. This "flexibility" means more than a general willingness to conserve energy. In this context, flexibility is the ability to reduce, shift, stage, or relocate electricity demand in response to grid conditions without shutting down the facility as a whole. Some computing tasks are latency-sensitive and difficult to interrupt, but other workloads can be delayed, throttled, moved across regions, or scheduled for periods when power is less constrained. Cooling systems may also provide short-term flexibility through thermal storage, pre-cooling, or chilled-water management. These features can make data centers more operationally flexible than many conventional industrial loads, especially when the operator controls geographically distributed computing capacity and sophisticated energy-management systems.

Data centers may become grid assets rather than only grid burdens, but only if their capabilities are converted into enforceable operating commitments. Demand response is an operational form of flexibility in which a customer temporarily reduces or shifts grid-supplied electricity use in response to a utility or grid-operator signal under defined notice, duration, compensation, and performance terms. Large facilities often include uninterruptible power supplies, batteries, backup generators, and sometimes other on-site generation or storage systems. Those resources can potentially help during constrained periods by reducing net load, supporting staged energization, or coordinating with demand-response programs. They can also create new risks if backup generation increases local emissions, if private reliability strategies conflict with system operations, or if promised flexibility is unavailable when the grid needs it. A tariff or service agreement must therefore define the quantity of flexible load, notice requirements, duration limits, compensation, penalties, telemetry, backup-generation rules, emissions constraints, and reliability obligations. Without those shared definitions, "flexibility" remains a general aspiration rather than an operational commitment.

Transparency is the administrative condition that makes those commitments usable. Utilities, regulators, and affected communities cannot evaluate a large-load project if essential facts about expected energization dates, peak demand, firm and interruptible load, duplicate interconnection requests, backup-power configuration, cooling design, water use, and emissions are unavailable or treated as purely private business information. An interconnection request is a developer's formal request for a utility to study and prepare to serve a specified amount of load at a particular location and time. Duplicate or speculative requests can cause utilities to plan facilities for demand that never materializes. The resulting stranded infrastructure costs are the costs of generation, transmission, distribution, or substation investments built for an anticipated load that does not materialize, closes, or materially reduces its demand, leaving those costs to be recovered from other customers.

Industry initiatives illustrate why policy design increasingly needs this technical vocabulary. The Electric Power Research Institute's (EPRI's) DCFlex initiative brings utilities, grid operators, technology companies, and data center operators together to demonstrate how data centers can support grid reliability and faster interconnection when flexibility is defined in operational terms [7, 8]. Its workstreams on grid-informed data center design, utility programs, and interconnection practices show that cooperation requires reference architectures, shared definitions of flexible service, and communication mechanisms between utilities and operators. State policy can give those cooperative arrangements public objectives: reliability, affordability, decarbonization, transparency, and protection of affected communities.

3 Case Selection and Analytical Framework

The universe of state data center policy is broader than the five jurisdictions examined in detail here. Some states have adopted tax incentives, general energy programs, or early-stage proposals, while others have begun to incorporate data center growth into grid forecasts without creating a data-center-specific regulatory regime. This paper narrows that broader field to states whose policies create binding obligations or concrete implementation processes for the electricity, water, environmental, or siting effects of data center development.

3.1 Case Selection

Because the background problem spans electric reliability, utility-cost allocation, water, emissions, land use, and local accountability, this paper does not treat a general tax incentive or a single-purpose economic-development statute as sufficient for inclusion in the main comparison. The five states were selected because each has adopted a statewide law, budget provision, binding executive action, or regulatory order that:

(1) expressly applies to data centers or to a class of very large electric loads that clearly encompasses major data centers;
(2) materially addresses at least two of the following subjects:
   (a) grid planning and interconnection;
   (b) operational demand flexibility or emergency curtailment;
   (c) allocation or disclosure of electricity-system costs;
   (d) energy sourcing, energy efficiency, or emissions;
   (e) water availability, water efficiency, or water-use reporting; and
   (f) environmental review, siting, or community impacts;
(3) imposes project-level obligations or creates a concrete implementation process, rather than merely offering a general tax incentive; and
(4) has statewide significance, even if implementation is delegated to utilities, environmental agencies, or local governments.

The method combines three forms of analysis. First, it reviews enacted statutes, budget provisions, executive orders, and regulatory proceedings to identify binding obligations or concrete implementation processes. Second, it compares those instruments against recurring regulation concerns: interconnection readiness, large-load tariffs, cost allocation, minimum load commitments, staged energization, demand response, emergency curtailment, and stranded-cost risk. Third, it evaluates whether each policy is technically responsive to data center operations, including load flexibility, backup-power coordination, clean-energy alignment, cooling-water requirements, and community resource constraints. The goal is not simply to catalogue laws, but to identify which legal designs create usable interfaces among public regulators, utilities, grid operators, data center developers, and host communities.

The five selected states satisfy these criteria through the instruments summarized in Table 1.

Several qualifications are necessary. Texas Senate Bill 6 applies to large electric loads generally, rather than exclusively to data centers, but it is included because modern data centers are central to the large-load problem in ERCOT. New York is included provisionally because its current model is precautionary and transitional while permanent standards remain under development. Virginia is included as a package rather than as a single comprehensive statute because its 2026 framework is distributed across utility, tax, water, air-emissions, and budget provisions. Appendix A discusses California and Illinois, which illustrate important adjacent policy activity but do not yet supply the same kind of binding, data-center-specific statewide framework used for the main comparison.

South Dakota Senate Bill 232, which would have imposed a one-year moratorium on statutorily defined hyperscale data centers, is not included as enacted law because the bill was tabled during the 2026 legislative session [17]. South Dakota's placement in this comparison rests on enacted Senate Bill 135.

3.2 Comparative Criteria

The comparison criteria translate the technical governance problem into legal questions. They ask whether a state can identify the relevant facilities, require credible information before grid investments are made, allocate costs fairly, convert latent flexibility into operational commitments, and address the water, emissions, siting, and implementation consequences of data center development. The same eight criteria organize the comparative analysis in Section 4.

- **C1: Facility scope and triggering thresholds.** This criterion asks which facilities are covered, whether the law is data-center-specific or technology-neutral, and whether the threshold is based on electric demand, water use, building size, capital investment, or another measure.
- **C2: Grid planning and interconnection.** This criterion considers interconnection studies, disclosure of project maturity, treatment of speculative or duplicate requests, transmission planning, staged energization, minimum commitments, and the allocation of upgrade obligations before a facility connects.
- **C3: Demand flexibility and emergency operations.** This criterion considers whether a facility may or must curtail load, participate in demand response, deploy on-site resources, accept non-firm service during stressed grid conditions, or provide telemetry and communications sufficient for grid operators to rely on the flexibility.
- **C4: Electricity costs and transparency.** This criterion includes protection of other customers from cross-subsidies and stranded costs, special customer classes or tariffs, direct charges or taxes, and measurement or reporting of data center electricity consumption.
- **C5: Energy sourcing, efficiency, and emissions.** This criterion considers clean-energy procurement, energy-efficiency requirements, sustainable-design conditions, and emissions from on-site or backup generation.
- **C6: Water availability, efficiency, and transparency.** This criterion considers advance water-supply review, withdrawal permitting, cooling-technology scrutiny, conservation and reuse requirements, public reporting, and authority to limit consumption.
- **C7: Enforcement and implementation maturity.** This criterion distinguishes immediately enforceable requirements from programs that depend on future rulemaking,

Table 1: Principal State Instruments and Covered Facilities

State	Principal instrument and date	Principal facilities targeted	Basis for selection
Texas	Senate Bill 6, signed and effective June 20, 2025 [20, 21].	New or expanded single-site large loads in the Electric Reliability Council of Texas (ERCOT) region above a threshold established by the Public Utility Commission of Texas, presumptively 75 MW. Several provisions also apply more broadly to new transmission-voltage customers.	Mandatory interconnection screening, customer financial commitments, duplicate-request disclosure, transmission planning, emergency curtailment, firm load-shed protocols, and review of transmission-cost allocation.
Minnesota	Chapter 12, H.F. 16, enacted during the 2025 First Special Session and signed June 14, 2025 [14].	For the principal energy provisions, facilities designed for a load of at least 100 MW whose primary purpose is digital-data storage, management, or processing. Separate triggers apply to water consumption and tax-incentive eligibility.	Integrated treatment of large-customer rates, clean-energy procurement, stranded-cost protection, water availability, conservation and reuse, environmental permitting, energy-related fees, and sustainable-design certification.
South Dakota	Senate Bill 135, signed March 24, 2026 [18].	Data centers operating as centralized facilities for electronic-data storage, management, dissemination, or processing with peak electric demand of at least 10 MW.	Full reimbursement of attributable utility costs, protection against utility shortages and cross-subsidies, advance water-demand evaluation, public water-use reporting, authority to impose water limits, and express preservation of local regulatory authority.
New York	Executive Order 62, issued July 14, 2026, together with Public Service Commission Case 26-E-0045, initiated February 12, 2026 [15, 19].	Under Executive Order 62, new hyperscale data centers capable of consuming at least 50 MW, subject to exclusions for certain manufacturing, research, educational, and medical facilities. The Public Service Commission (PSC) proceeding addresses large-load interconnection more generally.	Temporary suspension of covered environmental permits, a statewide generic environmental-impact review, and development of new rules for interconnection, cost allocation, demand response, water use, and community effects.
Virginia	A 2026 package signed June 30, 2026, including House Bill (H.B.) 284/Senate Bill (S.B.) 371, H.B. 892, H.B. 1191, H.B. 496, and H.B. 507, together with data center provisions in the 2026–2028 budget [4, 23–27].	The demand-flexibility program targets high-energy-demand customers with at least 25 MW of contracted or measured load and a load factor of at least 75 percent. The electricity tax and environmental provisions use separate data-center-specific definitions and triggers.	Voluntary large-load flexibility, improved load forecasting, ratepayer protection, a direct electricity-consumption tax, water-use reporting, water-efficient cooling requirements in sensitive areas, and stricter backup-generator emissions controls.

utility filings, agency studies, or temporary executive action.

The comparison evaluates the legal mechanisms adopted by each state. It does not attempt to measure actual compliance, the amount of flexible load enrolled, or the environmental performance of individual facilities.

4 Comparative Analysis by Criterion

The criteria examined below are analytically distinct but operationally connected. Interconnection rules shape infrastructure planning; cost-allocation rules determine who bears the resulting financial risk; flexibility provisions govern operation after energization; and energy, emissions, water, and siting requirements address effects that electric-service rules alone cannot resolve. The criterion-by-criterion analysis identifies the strengths and limitations of each instrument. Section 5 then considers what their distribution across states reveals about the coherence and completeness of the broader frameworks.

4.1 Facility Scope and Triggering Thresholds

Texas. Texas Senate Bill 6 is not formally limited to data centers. Its principal interconnection provisions apply to a new or expanded load at a single site in ERCOT when the load exceeds a threshold selected by the Public Utility Commission of Texas. The statute establishes 75 MW as the presumptive threshold, while allowing the Commission to adopt a lower threshold if necessary for reliability [20]. This design captures hyperscale data centers but also reaches other exceptionally large industrial or commercial loads.

The statute also applies certain operational requirements more broadly. For example, new transmission-voltage customers connecting after December 31, 2025, generally must maintain a firm load-shed protocol unless they qualify as protected critical loads. Texas thus uses multiple overlapping triggers rather than a single statutory definition of "data center."

Minnesota. Minnesota's principal energy-law definition covers a facility designed for a load of at least 100 MW whose primary purpose is the storage, management, or processing of digital data.

The definition includes associated backup generation, power distribution, environmental controls, cooling, and security systems [14]. However, the law employs separate thresholds for different obligations. Special water review applies when a covered data center proposes more than 100 million gallons per year of consumptive use and requires a new or amended appropriation permit. The tax-incentive provisions use a separate definition of a "qualified large-scale data center," including a minimum building-size and investment test. This layered approach allows the state to target different externalities using different measures, but it also makes the framework more complex.

South Dakota. South Dakota Senate Bill 135 applies to data centers, defined as centralized facilities used for electronic-data storage, management, dissemination, and processing, with peak electric demand of at least 10 MW [18]. This is the lowest explicit data center electric-demand threshold among the five states. It potentially reaches regional, enterprise, and co-location facilities that would fall below the Texas, Minnesota, or New York thresholds.

New York. Executive Order 62 applies to covered data centers capable of consuming at least 50 MW [19]. It focuses on facilities housing servers and related information-technology equipment and providing continuous cloud-computing, storage, content-delivery, or internal computing services.

The order excludes specified facilities whose primary purposes involve manufacturing, quantum or biomedical research, accredited education, certain state-supported research activities, or medical care. New York therefore uses both a quantitative threshold and a functional test. The separate Public Service Commission proceeding may ultimately produce interconnection rules that apply to a broader class of large loads [15].

Virginia. Virginia uses different thresholds for different policies. H.B. 284 and S.B. 371 define a "high-energy-demand customer" by reference to a contracted or measured load of at least 25 MW and an actual load factor of at least 75 percent [24]. This definition is not limited to data centers, although it captures their typically large and steady consumption.

The budget's electricity-consumption tax employs a data-center-specific definition that reaches facilities using at least 1 MW of qualifying power and cooling equipment. Water-efficiency and generator-emissions provisions use separate project, geographic, and permitting triggers [4, 26]. Virginia consequently reaches smaller data centers for taxation and environmental purposes than it does for high-load demand-flexibility programs.

Comparative Finding.

South Dakota has the broadest reach by electric-demand threshold, beginning at 10 MW. Virginia follows with a 25 MW threshold for its voluntary high-demand flexibility program, although some data-center-specific Virginia provisions reach smaller facilities. New York's principal threshold is 50 MW, Texas presumptively begins at 75 MW, and Minnesota's core energy definition begins at 100 MW.

Minnesota and Virginia make the most extensive use of multiple, policy-specific triggers. Texas instead uses a technology-neutral large-load threshold, while South Dakota and New York more directly identify data centers as a distinct regulated class.

4.2 Grid Planning and Interconnection

Texas. Texas has the most detailed enacted grid-interconnection framework. Senate Bill 6 requires a covered customer to contribute to the cost of interconnection studies and facilities. It directs the Public Utility Commission to establish a flat initial study fee of at least $100,000 and to require evidence of site control and uniform financial commitments before an application advances [20].

Applicants must disclose duplicate or substantially similar requests submitted elsewhere. This provision addresses the risk that utilities and ERCOT will count several speculative requests for the same project, overstate future demand, and build unnecessary infrastructure.

The law also requires forecast large loads to be incorporated into transmission planning and resource-adequacy assessments. For co-located load and generation arrangements, regulators may impose conditions concerning curtailment, generator availability, and responsibility for underused or stranded transmission facilities.

Minnesota. Minnesota does not establish a Texas-style standardized queue, study fee, or duplicate-request process. Its approach is instead centered on utility rate classification and resource responsibility. By December 15, 2026, the Public Utilities Commission must establish a very-large-customer class or subclass for each regulated utility [14]. Tariffs or special contracts for those customers must assign the costs of serving the class to that class and protect other customers from stranded costs. Minnesota also coordinates state permitting through the Business First Stop program, which can streamline agency review but is not a substitute for technical electric-interconnection rules.

South Dakota. South Dakota Senate Bill 135 does not create a detailed technical interconnection process. Instead, each electricity provider serving a covered data center must establish separate service terms requiring the facility to reimburse all costs fairly attributable to its service demand and consumption, including costs remaining if the facility departs or materially reduces load [18].

New York. New York's grid framework remains under development. In February 2026, the Public Service Commission initiated Case 26-E-0045 to examine interconnection reforms for large loads [15]. Executive Order 62 subsequently directed the Department of Public Service to establish a Data Center Interconnection Working Group and consider a beneficiary-pays approach to infrastructure costs, reforms to transmission and distribution practices, special service classifications, demand response, and protection against speculative loads [19]. However, the permanent standards had not beed finalized as of the writing of this paper.

Virginia. Virginia's 2026 legislation emphasizes more accurate forecasting and customer-funded infrastructure rather than a single standardized interconnection process. H.B. 892 seeks to improve forecasts of future electricity demand and reduce the risk that overstated load projections lead to unnecessary costs [27]. H.B. 1191 permits high-energy-use customers to support or invest in new energy infrastructure while protecting other ratepayers from associated costs [23].

Virginia's approach is integrated with utility-specific large-load tariffs and State Corporation Commission proceedings. It is therefore more decentralized than Texas's statutory ERCOT framework and relies more heavily on regulated-utility planning and rate cases.

Comparative Finding.

Texas is the clear leader in binding interconnection discipline. It directly addresses study costs, project maturity, duplicate requests, transmission planning, co-location, and stranded infrastructure.

New York is considering many of the same subjects but has not completed its rules. Virginia and Minnesota focus more on forecasting, tariffs, class responsibility, and customer-funded infrastructure. South Dakota provides strong protection against unrecovered costs but comparatively little procedural detail concerning how large loads enter and move through an interconnection queue.

4.3 Demand Flexibility and Emergency Operations

Texas. Texas provides the strongest mandatory emergency authority. A covered customer that has on-site backup generation capable of serving at least half of its demand must disclose that resource. After available market-based reliability tools have been used, ERCOT may require deployment of qualifying backup generation or reduction of the customer's grid demand during an emergency [20].

The statute also requires firm load-shed protocols for many new transmission-voltage customers. It directs development of a competitive reliability service under which large loads of at least 75 MW may receive advance notice and reduce demand during specified reliability events. Once an emergency begins, required curtailment may continue for the duration of the event.

The Texas framework therefore combines compensated or competitively procured flexibility with mandatory emergency control.

Minnesota. Minnesota authorizes an optional clean-energy-and-capacity tariff for commercial and industrial customers. A participating customer may contract for incremental clean-energy resources and associated capacity and must pay its proportional costs [14].

The statute does not create an ERCOT-style emergency-disconnection regime for data centers. Demand flexibility may be incorporated into utility tariffs, special contracts, or voluntary arrangements, but mandatory operational curtailment is not the central feature of the 2025 law.

South Dakota. South Dakota Senate Bill 135 directs attention to whether serving a data center could create an electricity shortage and requires utility terms that protect existing customers. The state, therefore, addresses resource adequacy principally through service conditions and cost responsibility rather than direct operational dispatch of the data center load.

New York. Executive Order 62 instructs state agencies to evaluate demand response, distributed resources, storage, and other mechanisms that could mitigate the grid effects of hyperscale data centers [19]. New York already has general utility demand-response programs, but Executive Order 62 does not itself establish a detailed mandatory curtailment protocol for covered facilities.

Virginia. H.B. 284 and S.B. 371 require Dominion Energy and Appalachian Power to develop voluntary demand-flexibility programs for high-energy-demand customers [24]. The programs are intended to allow customers such as large data centers to shift, reduce, or otherwise manage consumption in response to grid needs and price conditions.

Virginia's model is therefore explicitly voluntary. It seeks to turn large loads into compensated or contractually managed flexibility resources without granting the grid operator the same broad emergency-disconnection authority established in Texas.

Comparative Finding.

Texas provides the strongest direct operational control because it combines emergency authority, firm load-shed protocols, backup-generation disclosure, and a reliability demand-reduction service. Virginia has the clearest enacted voluntary large-load-flexibility mandate. Minnesota provides an optional procurement mechanism but less direct operational control. New York is developing flexibility mechanisms, while South Dakota focuses on adequacy and customer protection rather than dispatchable demand response.

Flexibility must be specified before it can be trusted. DCFlex and related utility experiments indicate that data center flexibility can include workload shifting, staged energization, temporary demand reduction, thermal management, battery dispatch, or carefully governed use of backup generation [7, 9]. Those capabilities are not interchangeable. A grid operator needs to know how much load can move, how quickly it can respond, how long the response can last, how often it can be called, and what emissions or reliability constraints apply if backup resources are used. Texas most converts this idea into law, while Virginia converts it into voluntary program.

4.4 Electricity Costs and Transparency

Texas. Cost allocation is a central purpose of Senate Bill 6. Covered customers must contribute to interconnection costs and demonstrate financial commitment. The statute also directs regulators to evaluate the methodology used to allocate transmission-system costs and to amend that methodology by the end of 2026 where appropriate [20].

Texas's model is strong on protecting other ratepayers from infrastructure costs and speculative projects. It is less strong on public disclosure of actual facility-level electricity consumption, energy efficiency, or monthly electricity expenditures.

Minnesota. Minnesota requires utilities to establish a very-large-customer class or subclass and to allocate the costs of service to that class. Tariffs and special agreements must protect other customers against stranded costs [14].

The law also imposes an annual fee on qualified large-scale data centers according to forecast peak demand. The statutory fee ranges from $2 million for facilities in the lowest covered peak-demand band to $5 million for facilities at or above 750 MW. Revenue supports energy-conservation and related programs.

Minnesota additionally ended or narrowed portions of the electricity sales-tax treatment previously available to large data centers. These provisions make the financial consequences of large electricity demand more explicit, although the statute does not require comprehensive public publication of each facility's actual power-usage profile.

South Dakota. South Dakota requires a data center's electricity provider to establish separate terms under which the data center reimburses all costs fairly attributable to its service. The obligation includes costs that would otherwise become stranded if the facility closes, departs, or materially reduces its demand [18].

This is a strong anti-cross-subsidy rule. Its simplicity may make the policy easier to state than to implement, however, because attribution of shared generation, transmission, and distribution costs may still require utility and regulatory judgment.

New York. Executive Order 62 directs agencies to develop a beneficiary-pays model and consider special service classifications for data centers [19]. It also calls for consideration of financial mechanisms to address speculative loads and stranded infrastructure, potentially including upfront capital commitments or risk-pooling arrangements.

The Public Service Commission proceeding is intended to determine how large-load costs should be assigned and how existing customers should be protected [15]. These mechanisms are not yet complete, so New York's cost framework is currently less certain than those in Texas, Minnesota, or South Dakota.

Virginia. Virginia has adopted the most direct tax-based measurement of data center electricity consumption. The 2026–2028 budget imposes a tax of $0.011 per kilowatt-hour (kWh) on electricity consumed by covered data center operators beginning July 1, 2026, with the initial rate applying through June 30, 2028 [4].

The tax applies to both utility-supplied and self-supplied electricity. Utilities must separately identify the tax on customer invoices, while operators using self-supplied electricity must report that consumption to the Department of Environmental Quality. These provisions create administrative visibility into electricity use, although tax information may not necessarily be released publicly at facility-level granularity.

Virginia also uses improved forecasting, customer-funded infrastructure, and utility tariffs to reduce cost shifting [23, 27].

> **Comparative Finding.**
> Texas provides the most detailed protection against speculative interconnection and stranded grid investments. Minnesota and South Dakota establish the clearest class- or customer-based principles requiring data centers to pay attributable system costs. Virginia provides the strongest direct mechanism for measuring and charging electricity consumption through its per-kWhtax. New York is developing a beneficiary-pays model but has not yet finalized it.

4.5 Energy Sourcing, Efficiency, and Emissions

Texas. Senate Bill 6 does not require covered customers to procure renewable or carbon-free electricity, meet a power-usage-effectiveness target, or obtain an energy-efficiency certification. Its treatment of on-site generation is primarily reliability-oriented. The statute makes clear that deployment of backup generation during emergencies remains subject to applicable environmental and emissions requirements [20]. It does not itself create stricter emissions limits for diesel generators.

Minnesota. Minnesota's optional clean-energy-and-capacity tariff allows large customers to fund new clean generation and the capacity needed to serve their loads. Participating customers bear the proportional cost, thereby reducing the risk that incremental clean resources are subsidized by other customers [14]. The law also requires qualifying incentive recipients to obtain an approved sustainable-design or energy-management certification within the statutory period. Eligible systems include recognized building, infrastructure, and energy-management standards. Failure to obtain the required certification can result in tax recapture.

South Dakota. Senate Bill 135 does not establish a clean-energy procurement requirement, energy-efficiency target, or backup-generator emissions standard specific to data centers [18]. Its energy provisions are concerned mainly with adequacy, attributable cost, and protection of existing utility customers.

New York. Executive Order 62 directs the generic environmental review to evaluate energy demand, air quality, greenhouse-gas consequences, and the potential role of clean generation, storage, distributed resources, and demand response [19]. These subjects may form the basis of future permitting standards or conditions.

The current order does not itself impose a finalized renewable-energy percentage, efficiency metric, or generator-emissions limit. New York's position is therefore environmentally ambitious but not yet operationally complete.

Virginia. Virginia has enacted a direct emissions requirement for data center backup generation. Under H.B. 507, qualifying air-permit applications submitted on or after July 1, 2026, must meet emissions controls equivalent to federal Tier 4 standards or a stricter applicable standard [26].

Virginia's electricity-consumption tax also creates a financial signal tied directly to energy use. The state has not, however, adopted a data-center-specific renewable-energy matching requirement or statewide numerical power-usage-effectiveness standard.

> **Comparative Finding.**
> Minnesota has the strongest enacted framework for clean-energy procurement and sustainable design. Virginia has the clearest data-center-specific backup-generator emissions requirement. New York is conducting the broadest prospective review of energy and emissions effects, but the resulting standards remain pending. Texas and South Dakota concentrate on system reliability and cost rather than source energy, efficiency, or emissions.

4.6 Water Availability, Efficiency, and Transparency

Water policy is where the link between data center engineering and community impact is most visible. Cooling towers, chilled-water systems, air-side economization, closed-loop designs, reclaimed water, and direct evaporative systems impose different burdens on electric consumption, potable-water demand, wastewater discharge, and heat rejection. A technically coherent policy therefore should not ask only whether a facility has a legal water source. It should ask whether the proposed cooling architecture is appropriate for local water conditions, whether alternatives have been evaluated, whether actual consumption will be disclosed, and whether claimed conservation or replenishment measures correspond to measured withdrawals and consumptive use.

Texas. Texas laws contain no substantive water-use provision specific to data centers. However, data centers remain subject to generally applicable state and local water rules.

Minnesota. Minnesota has the most technically detailed water-permitting framework. A data center proposing more than 100 million gallons per year of consumptive use and requiring a new or amended appropriation permit is subject to a special preapplication evaluation [14].

The Department of Natural Resources may request projected maximum-day, maximum-month, and annual water consumption, the proposed source, and water-quality or temperature requirements. It must evaluate potential water-availability constraints and may coordinate with health, agriculture, and pollution-control agencies.

Permit review may consider:

- conservation and efficient use;
- recycling and reuse;
- reclaimed-water supplies;
- closed-loop cooling;
- watershed restoration or replenishment;
- conflicts with existing users or ecosystems; and
- additional aquifer testing or related technical evidence.

The requirements can also apply when an existing permit holder proposes to supply more than the statutory volume to a data center.

South Dakota. Before beginning operation, a covered data center must provide projected water consumption to each local water provider from which it expects to receive water. Each provider must determine whether the projected demand is compatible with available supply [18].

The state Board of Water Management may impose consumption limits after accounting for residential needs and essential public services. The law also requires semiannual reports identifying average water use and certifying compliance. Water-use information submitted under the law is public.

New York. Executive Order 62 directs the statewide generic environmental-impact review to consider water quantity, water quality, and associated environmental effects [19]. It also directs the Department of Environmental Conservation to determine whether new or amended water-withdrawal regulations, policies, reporting requirements, or guidance are needed for data centers.

Virginia. Virginia has adopted both water-efficiency and reporting measures. The 2026–2028 budget directs the Department of Environmental Quality to identify Cooling Water Scarcity Areas. For covered projects in those areas, data centers must demonstrate that they have minimized cooling-water use and employed the best available water-efficient technologies [4].

The budget identifies approaches such as:

- air-based cooling;
- closed-loop systems;
- recycled water;
- stormwater reuse; and
- nonpotable reclaimed water.

Related requirements apply to specified new facilities in the Eastern Virginia Groundwater Management Area. Direct evaporative cooling is restricted unless used in conjunction with an approved primary cooling technology. H.B. 496 separately improves water-withdrawal and water-use reporting by requiring water suppliers to disaggregate quantities supplied to data centers and other specified customer categories [25].

Comparative Finding.

Minnesota has the most complete advance water-permitting process and the broadest express consideration of conservation, reuse, cooling technology, ecosystems, and competing users. Virginia has the strongest technology-oriented cooling rules in designated water-sensitive areas. South Dakota has the strongest combination of advance local supply review, public facility reporting, and authority to impose consumption limits.

New York is developing a potentially broad framework but has not finalized it. Texas Senate Bill 6 does not materially regulate water use.

4.7 Enforcement and Implementation Maturity

Texas. Texas Senate Bill 6 is enacted and effective, and its principal statutory requirements are binding. Several details nevertheless depend on continuing Public Utility Commission and ERCOT implementation, including the precise large-load threshold, interconnection procedures, reliability-service design, and transmission-cost methodology [20].

Minnesota. Minnesota's water provisions, fee structure, tax changes, and agency responsibilities are enacted. The Public Utilities Commission must still establish very-large-customer classifications and related tariff structures, with an initial statutory deadline of December 15, 2026 [14].

South Dakota. South Dakota Senate Bill 135 establishes direct duties for data centers, electricity providers, water providers, and the Board of Water Management [18]. Its cost-allocation and reporting rules are comparatively straightforward, although implementation may require utility agreements and administrative decisions concerning attributable costs and appropriate water limits.

New York. New York has the least mature permanent framework. Executive Order 62 currently restrains permitting and directs agency action, but many of the ultimate interconnection, water, cost-allocation, and environmental standards depend on the generic environmental-impact statement, the Public Service Commission proceeding, and subsequent rulemaking [15, 19].

Virginia. Virginia's electricity tax, backup-generator emissions law, water-reporting law, budget provisions, and utility-program mandates have been enacted [4, 24–26]. Some elements remain prospective: utilities must design and submit demand-flexibility programs, the Department of Environmental Quality must designate water-scarcity areas, and agencies must develop implementation procedures.

Comparative Finding.

Texas currently has the most mature large-load grid statute. South Dakota has a relatively direct and immediately understandable framework for cost and water accountability. Minnesota and Virginia have enacted broader packages but require significant regulatory and agency implementation. New York's current intervention is binding, but its permanent standards remain the least complete.

5 Comparative Findings

Table 2 summarizes the relative strengths of the five frameworks. The ratings are qualitative and compare only the principal instruments discussed in this paper. "Leading" indicates the strongest or most complete approach among the five states; "strong" indicates a meaningful and enforceable framework; "moderate" indicates material but narrower requirements; "emerging" indicates that policy development is substantial but incomplete; and "limited" indicates that the principal state instrument provides little direct regulation of that subject.

No single state provides the leading model across every criterion:

- **Texas** provides the strongest model for large-load interconnection, project-readiness screening, emergency grid operations, and protection against stranded electric-system investments.
- **Minnesota** provides the most comprehensive integrated framework, particularly where energy procurement, water availability, water conservation, sustainable design, and coordinated permitting are considered together.
- **South Dakota** provides the broadest coverage by demand threshold and a clear model for direct cost responsibility, local water-supply review, public water-use reporting, and preservation of local land-use authority.
- **New York** provides the strongest precautionary and cumulative environmental-review model, but its placement in the main comparison is provisional because the permanent rules contemplated by its executive order and regulatory proceeding remain unfinished.
- **Virginia** provides the strongest hybrid model: voluntary grid flexibility, direct taxation and measurement of electricity consumption, improved load forecasting, water-efficient cooling in sensitive areas, water-use reporting, and generator-emissions controls.

This analysis shows that breadth and technical coherence are not the same. Strong interconnection rules can protect the grid while leaving water use or emissions largely unaddressed; environmental review can identify cumulative impacts without establishing mature operating or cost-allocation rules; and broad substantive requirements may have limited immediate effect when implementation depends on future tariffs, programs, or agency standards. The criteria are therefore complementary rather than interchangeable. A coherent framework must connect project-readiness screening and infrastructure-cost responsibility with operational flexibility, environmental performance, resource transparency, and enforceable implementation.

6 Policy Implications and Conclusion

The comparison demonstrates that data center regulation is becoming a distinct form of infrastructure governance. The issue is not simply that data centers are large. It is that their technical characteristics–high load factor, rapid energization demands, strict reliability requirements, backup-power systems, controllable computing workloads, and cooling-system tradeoffs–create public risks that conventional commercial-service rules do not address well. Regulation is essential not because it substitutes for cooperation between utilities and data center operators, but because it gives that cooperation clear public objectives.

Across the five states, the strongest emerging policies share four features. First, they formalize the utility–operator interface by requiring credible interconnection information, project-readiness evidence, minimum commitments, staged service options, and defined flexibility obligations. Second, they protect households and smaller businesses from cross-subsidizing speculative or underused infrastructure through attributable-cost rules, customer classes, taxes, or beneficiary-pays principles. Third, they connect electricity service to environmental performance through clean-energy procurement, emissions controls, or sustainable-design requirements. Fourth, they treat cooling and water use as resource-governance questions rather than purely private design choices.

No state yet combines all of these elements. The comparative findings instead suggest that the strongest model would be modular: it would borrow different tools from different states and connect them through a common theory of large-load governance.

A comprehensive model statute could combine the strongest features of all five states:

(1) Texas-style project-readiness screening, interconnection discipline, and emergency flexibility;
(2) Minnesota-style clean-capacity procurement and detailed water permitting;
(3) South Dakota-style public water reporting and local-control protections;
(4) New York-style cumulative environmental and community-impact review; and
(5) Virginia-style electricity-consumption measurement, water-efficient cooling, and backup-generator emissions standards.

Such a framework would distinguish between fully firm and flexible electric service, require large facilities to internalize the infrastructure risks they create, provide transparent information about electricity and water use, and regulate environmental effects without treating every data center as operationally or environmentally identical. The broader lesson is that data center policy should not be framed as a binary conflict between digital-infrastructure growth and community protection. A better frame is technical and institutional alignment: rules that make load behavior, cost responsibility, clean-energy obligations, cooling choices, and local resource impacts legible before facilities are built. Those rules remain fragmented across jurisdictions, but the elements are now visible enough to guide federal, regional, or multi-state coordination. The goal is not to deter data centers, but to establish the conditions under which they can become good infrastructure neighbors.

A Other State Activity Outside the Main Comparison

Several other states are responding to data center growth, but their current approaches either remain broader than data center policy, rely mainly on incentives, or have not yet produced binding data-center-specific obligations comparable to the five states selected for detailed comparison.

California illustrates a grid-planning and flexibility model rather than a data-center-specific regulatory model. The California Energy Commission reports that it does not permit data centers as such, but it does assess their effects on future electricity demand and has incorporated projected data center growth into its statewide

Table 2: Qualitative Comparison of the Five State Frameworks

Criterion	Texas	Minnesota	South Dakota	New York	Virginia
Facility scope and triggering thresholds	Moderate	Strong	Leading	Strong	Strong
Grid planning and interconnection	Leading	Moderate	Moderate	Emerging	Strong
Demand flexibility and emergency operations	Leading	Moderate	Limited	Emerging	Strong
Electricity costs and transparency	Strong	Strong	Strong	Emerging	Leading
Clean-energy sourcing and efficiency	Limited	Leading	Limited	Emerging	Moderate
Water availability, efficiency, and transparency	Limited	Leading	Strong	Emerging	Strong
Environmental and community review	Limited	Strong	Moderate	Leading/Interim	Strong
Implementation maturity	Leading	Strong/Emerging	Strong	Interim	Strong/Emerging

demand forecast using utility energization-application data [1]. The same CEC discussion states that California had more than 200 active data centers, that data centers accounted for about 1,000 MW or 2 percent of California Independent System Operator peak demand in early 2026, and that demand could reach 4,500 MW or 9 percent of peak demand by 2040 [1]. California also has general demand-flexibility tools that could apply to large customers, including the Demand Side Grid Support program, which offers incentives for load reduction and backup generation during extreme events, and the Public Utilities Commission's demand-flexibility rulemaking, which treats demand response and load flexibility as resources for integrated resource planning [2, 3]. These programs are important because they show how a state can use forecasting and systemwide flexibility policy to manage new loads, but they do not by themselves impose the kind of data-center-specific interconnection, cost-allocation, water, or siting regime used as the basis for this paper's main comparison.

Illinois illustrates a different category: an incentive-centered approach with some emerging but still incomplete resource-protection proposals. The Illinois Department of Commerce and Economic Opportunity describes the state data center investment program as providing qualifying data centers exemptions from state and local taxes and a construction-employment tax credit, with eligibility tied to at least $250 million in capital investment, at least 20 new full-time or full-time-equivalent jobs, wage requirements, and either carbon neutrality or certification under a green-building standard [11]. The same agency states that, pursuant to the Governor's June 5, 2026 directive, it would stop processing applications for that incentive program as of July 1, 2026 [11]. Illinois has also adopted the Clean and Reliable Grid Affordability Act, a broad energy law aimed at reliability, affordability, storage procurement, energy efficiency, and long-term planning, but the Illinois Power Agency describes that law as a general energy framework rather than a data-center-specific statute [13]. A more targeted proposal, Senate Bill 4004, would create a Data Center Water Transparency and Aquifer Protection Act restricting data center withdrawals from the Mahomet Aquifer and limiting nondisclosure agreements that conceal data center water use, but the bill was re-referred to Assignments on May 22, 2026 and had not become enacted law [12]. Illinois therefore shows meaningful policy movement, but the binding statewide framework remains weaker and less data-center-specific than the principal instruments compared in this paper.

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